

Irodov Problem 1.13

Rajarshi Dasgupta

September 11, 2026

1.13 Point A moves uniformly with velocity v so that the vector $\vec{v}$ is continually ‘aimed’ at point B which in its turn moves rectilinearly and uniformly with velocity $u < v$. At the initial moment of time $\vec{v} \perp \vec{u}$ and the points are separated by a distance l . How soon will the points converge?

I want to discuss the solution of the above problem from ‘Problems in General Physics’ by I.E. Irodov [1]. Let T be the time taken for points A and B to converge. So, we need to find an expression for T in terms of the quantities u , v , and l .

1 School kid’s solution

Let r be the distance between the points A and B, and θ be the angle between the velocities $\vec{v}$ and $\vec{u}$. We consider the relative velocity of A with respect to B along AB and orthogonal to AB to obtain the following relation.

$$\dot{r} = -v + u \cos \theta \quad (1)$$

$$r\dot{\theta} = -u \sin \theta \quad (2)$$

First, we will eliminate $\dot{\theta}$ by differentiating equation 1 with respect to time and then substituting the expression for $\dot{\theta}$ from equation 2.

$$\begin{aligned} \ddot{r} &= -u\dot{\theta} \sin \theta \\ \implies r\ddot{r} &= u^2 \sin^2 \theta \end{aligned}$$

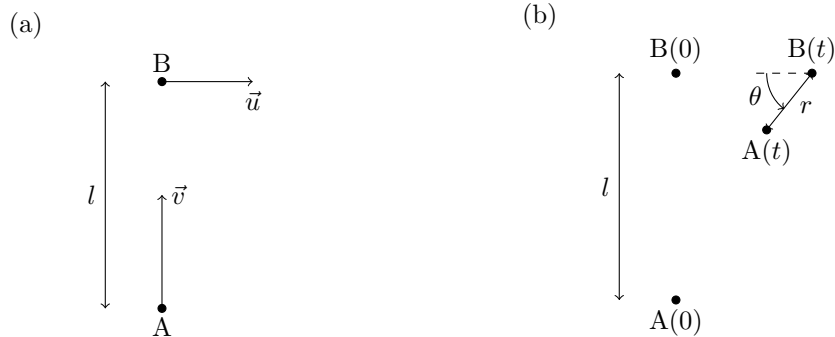


Figure 1: (a) Velocities of points A and B separated by a distance of l at the initial moment of time $t = 0$ (b) Velocities of points A and B separated by a distance of r at time $t \in (0, T)$.

Now, we will eliminate θ from the above expression by using the trigonometric identity $\cos^2 \theta + \sin^2 \theta = 1$, and equation 1.

$$\begin{aligned} r\ddot{r} &= u^2 \sin^2 \theta \\ &= u^2 (1 - \cos^2 \theta) \\ &= u^2 \left\{ 1 - \left[\frac{\dot{r} + v}{u} \right]^2 \right\} \end{aligned}$$

Simplifying the above expression we obtain the following second order differential equation.

$$r\ddot{r} + \dot{r}^2 + 2\dot{r}v + v^2 - u^2 = 0 \quad (3)$$

$$\implies \frac{d}{dt}(r\dot{r} + 2rv) + v^2 - u^2 = 0 \quad (4)$$

Now, integrating with respect to time from $t = 0$ to T we get the following expression.

$$\begin{aligned} &\int_{t=0}^{t=T} \frac{d}{dt}(r\dot{r} + 2rv)dt + \int_{t=0}^{t=T} (v^2 - u^2)dt = 0 \\ \implies &r(T)\dot{r}(T) + 2r(T)v - r(0)\dot{r}(0) - 2r(0)v + (v^2 - u^2)T = 0 \end{aligned}$$

We will use the fact that

$$\begin{aligned} r(0) &= l \\ \dot{r}(0) &= -v + u \cos \pi/2 && \text{(from equation 1)} \\ &= -v \\ r(T) &= 0 && \text{(since A and B converge at time T)} \end{aligned}$$

to get the final expression for T .

$$T = \frac{lv}{v^2 - u^2} \quad (5)$$

2 Numerical solution

Interestingly, the above analytical solution does not give us the trajectory of A. Given numerical values of u , v , and l , the trajectory of point A relative to B can be found by considering the ordinary differential equations 1 and 2 to formulate an initial value problem as $r(0) = l$ and $\theta(0) = \pi/2$. On applying the forward Euler's method we arrive at the following scheme to find the trajectory of A.

$$r_{n+1} = r_n + \Delta t(u \cos \theta - v) \quad (6)$$

$$\theta_{n+1} = \theta_n - \Delta t u \sin \theta \quad (7)$$

where Δt is the time step considered for the discretisation of time, that is,

$$r_n \approx r(n\Delta t)$$

$$\theta_n \approx \theta(n\Delta t)$$

Given below is a python function to simulate the motion of A relative to B.

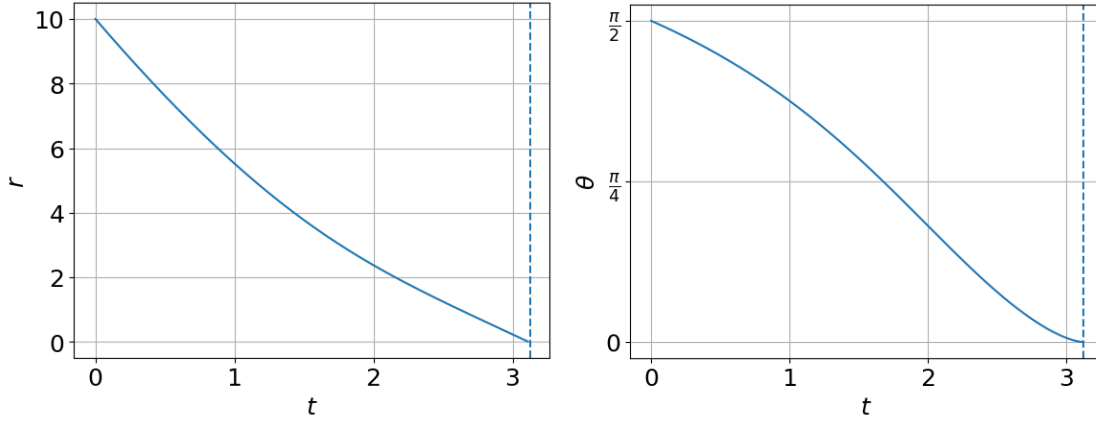


Figure 2: Simulation of r and θ for $(u, v, l) = (3, 5, 10)$, where r is the distance AB and θ the angle between the velocities of A and B. The dashed line indicates the time of contact T , according to the analytical solution given by equation 5, $T = lv/(v^2 - u^2)$.

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import numpy as np
import matplotlib.pyplot as plt

def numSol(u, v, l, dt, tolFrac = 1e-4, NMax = 100_000):
    # Initial Condition
    r = l
    th = np.pi/2

    rList = []
    thList = []

    tol = tolFrac * l
    # Forward Euler steps till r > tol
    stepCount = 0
    while r > tol and stepCount < NMax:
        stepCount += 1
        rList.append(r)
        thList.append(th)

        rNext = r + dt * (u*np.cos(th) - v)
        thNext = th - dt * u*np.sin(th) / r

        r, th = rNext, thNext

    # T_answer = stepCount * dt
    return stepCount, np.array(rList), np.array(thList)

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The consistency of the analytical result from equation 5 with the simulated relative position of A with respect to B for $(u, v, l) = (3, 5, 10)$ can be seen in figure 2. We observe this consistency for a wide range of problem parameters u , v and l .

3 Physics major's solution

Let us treat the time to collision T , as a function of r and θ . Now, given a trajectory $r(t), \theta(t)$, the rate of change of T with respect to t should be precisely -1.

$$\begin{aligned}\frac{dT}{dt} &= -1 \\ \implies \dot{r} \frac{\partial T}{\partial r} + \dot{\theta} \frac{\partial T}{\partial \theta} &= -1\end{aligned}$$

Now, using equations 1 and 2 we arrive at a partial differential equation for T .

$$\nabla T \cdot [(-v + u \cos \theta) \hat{e}_r - u \sin \theta \hat{e}_\theta] = -1 \quad (8)$$

Remember that in polar coordinates, $\nabla T = \hat{e}_r \partial T / \partial r + (1/r) \hat{e}_\theta \partial T / \partial \theta$. To pose this as a boundary value problem, we will use the known values of T along the x -axis.

$$T(r, \theta) = \begin{cases} \frac{r}{v-u}, & \text{for } r > 0, \theta = 0 \\ 0, & \text{for } r = 0 \\ \frac{r}{v+u}, & \text{for } r > 0, \theta = \pi \end{cases} \quad (9)$$

Let us consider the following form of the solution for the boundary value problem described by equations 8 and 9.

$$T(r, \theta) = R(r)A(\theta)$$

From the boundary values, equation 9, it follows that $R(r)A(0) = r/(v+u)$, hence we can choose $R(r) = r$ and absorb other constant factors in $A(0)$. This linearity of R can also be justified from equation 8 as $(-v + u \cos \theta) \hat{e}_r - u \sin \theta \hat{e}_\theta$ and the right hand side are both independent of r . Now, plugging the ansatz $T = rA(\theta)$ into the partial differential equation 8, we get a first order equation for A

$$\begin{aligned}(-v + u \cos \theta)A - u \sin \theta \frac{dA}{d\theta} &= -1 \\ \implies \frac{dA}{d\theta} + \frac{v - u \cos \theta}{u \sin \theta} A &= \frac{1}{u \sin \theta} \\ \implies \frac{d}{d\theta}(\mu A) &= \frac{\mu}{u \sin \theta}\end{aligned}$$

where

$$\mu(\theta) = \exp \int \frac{v - u \cos \theta}{u \sin \theta} d\theta \quad (10)$$

is the integrating factor. After finding μ , $A = \frac{1}{\mu} \int \frac{\mu}{u \sin \theta} d\theta$ and working out the arbitrary constants using equation 9, we get

$$A(\theta) = \frac{1}{v} \frac{1 + (u/v) \cos \theta}{1 - (u/v)^2} \quad (11)$$

$$T(r, \theta) = rA(\theta) = r \frac{v + u \cos \theta}{v^2 - u^2} \quad (12)$$

It can be verified, that this expression for T satisfies the boundary value problem defined by equations 8 and 9. From mathematical arguments of existence and uniqueness we can be assured that it is indeed the correct solution. The visualisation of T and A can be seen in figure 3.

So, for solution to the original problem, we plug $r = l$ and $\theta = \pi/2$ to get $T = lv/(v^2 - u^2)$ which is identical to equation 5.

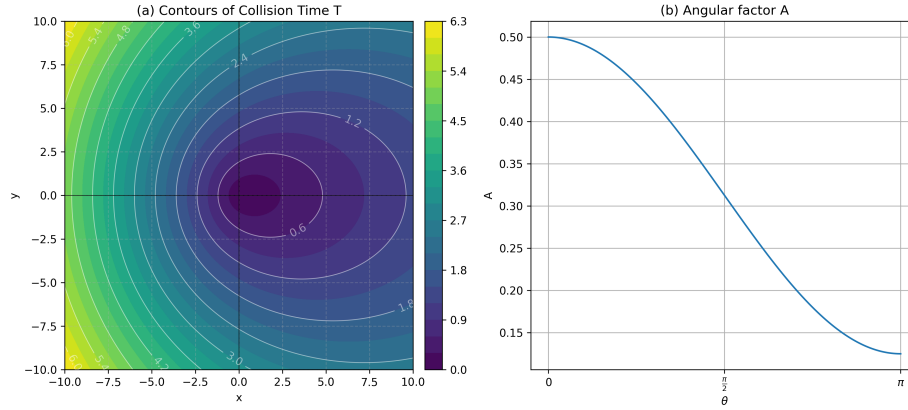


Figure 3: The coordinates r and θ are the distance between particles and the angle between their velocities respectively. Given a point (x, y) , $r = \sqrt{x^2 + y^2}$ and θ is the angle from the negative x -axis. (a) Contours of collision time $T(r, \theta) = rA(\theta)$ (b) Function A of θ given by equation 11

Discussion

We have a problem in two dimensions, and so, it is only natural that all aforementioned attempts involve the coupled first order ordinary differential equations 1 and 2. The frame of reference is set on particle B with polar bases, which moves rectilinearly and uniformly with respect to the ground. It is interesting to note that r and θ decrease monotonically with time, that is, $\dot{r}, \dot{\theta} < 0$. So, as the distance decreases with time, we can be assured of collision, $r_{\text{final}} = 0$. From equation 2, we can also discern the angle of impact $\theta_{\text{final}} = 0$ or π . In other words, collision is guaranteed (as $v > u$) and that it happens either from behind or head on. Interestingly enough, upon collision, the kinematics, or the rules of motion set by the problem, can not be followed further in time.

At first glance, it seems like we could determine the trajectory, $r(t), \theta(t)$ as closed form expressions of time t . Following the school kid's approach, we first arrive at a second order differential equation upon eliminating θ , and then with a stroke of luck, we find that it can be integrated once to decrease the order and arrive at equation 4. Before collision, when $t < T$ and $r > 0$, to find the closed form solution for $r(t)$, we are required to solve the following initial value problem.

$$\begin{aligned} r(0) &= l, \dot{r}(0) = -v + u \cos(\pi/2) = -v \\ r(t)\dot{r}(t) + 2vr(t) + (v^2 - u^2)t - lv &= 0 \end{aligned}$$

This may fall into the category of the Abel equation of the second kind, which as it turns out can not be solved in general. My knowledge regarding this is limited, so I went ahead to solve for the collision condition, where $t = T$ and $r(T) = 0$ simplify the expression and yield the answer, which is T expressed as function of u, v and l .

The other more complicated theoretical attempt involves treating the time to collision T as a field and transforming the problem to standard boundary value problem. This approach seems to involve dirty mathematical tricks of which a physics major's student may be acquainted with and not rely on lucky strikes.

Finally, the numerical solution, shedding some light on the trajectory, can found by applying the discrete time approximations from equations 6 7.

References

- [1] I. E. Irodov, *Problems in General Physics*, Mir Publishers, 1988.